

For a given plug, the radii r_w (gas-side wall radius), r_a , and r_b are collinear; they are taken perpendicular to the gas-side surface of the nozzle and extend to the centerline; T_a and T_b are internal wall temperatures measured with the thermocouples embedded in the plug. The thermal conductivity k is the arithmetic average of the values at T_a and T_b .

The gas-side wall temperatures determined from the different thermocouple combinations in each plug were generally within 1%. However, in determining the wall heat flux from Eq. (1), there were inconsistencies. If the center thermocouple and the one nearest the gas-side wall were used, the calculated wall heat flux was on the average about 10% higher than with the thermocouples nearest the gas-side and water-side walls. With a combination of the center thermocouple and the one nearest the gas-side wall, the total heat load was found to agree within 5% of that computed from the coolant flow rate and the coolant temperature rise; consequently, these two thermocouples were used to calculate the wall heat flux. The estimated errors resulting from the use of these thermocouples are discussed near the end of Appendix A.

The heat-transfer coefficient was computed by

$$h = \frac{q_w}{T_{aw} - T_w} \quad (2)$$

The adiabatic wall temperature T_{aw} was calculated from a recovery factor assumed equal to the 1/3 power of the stagnation temperature value of the Prandtl number:

$$\frac{T_{aw}}{T_t} = \frac{1 + Pr^{1/3} \frac{\gamma - 1}{2} M^2}{1 + \frac{\gamma - 1}{2} M^2}$$

The Mach number M was determined from the experimental static-to-stagnation pressure ratio p/p_t for isentropic flow ($\gamma = \text{const.}$) at the corresponding stagnation temperature value of γ for the products of combustion. The adiabatic wall temperatures so calculated differ only slightly from those corresponding to a recovery factor equal to 0.89. This value is based on measurements with air accelerated over a flat plate by a convergent opposite wall (Ref. 8) and by extrapolating wall temperatures to the zero heat-flux condition for air flow through a nozzle (Ref. 4). In both of these investigations, the recovery factor was found to be independent of pressure gradient. Actually, for the large differences between the stagnation and wall temperatures in the present results, the calculated heat-transfer coefficients are insensitive to the assumed recovery-factor dependence.

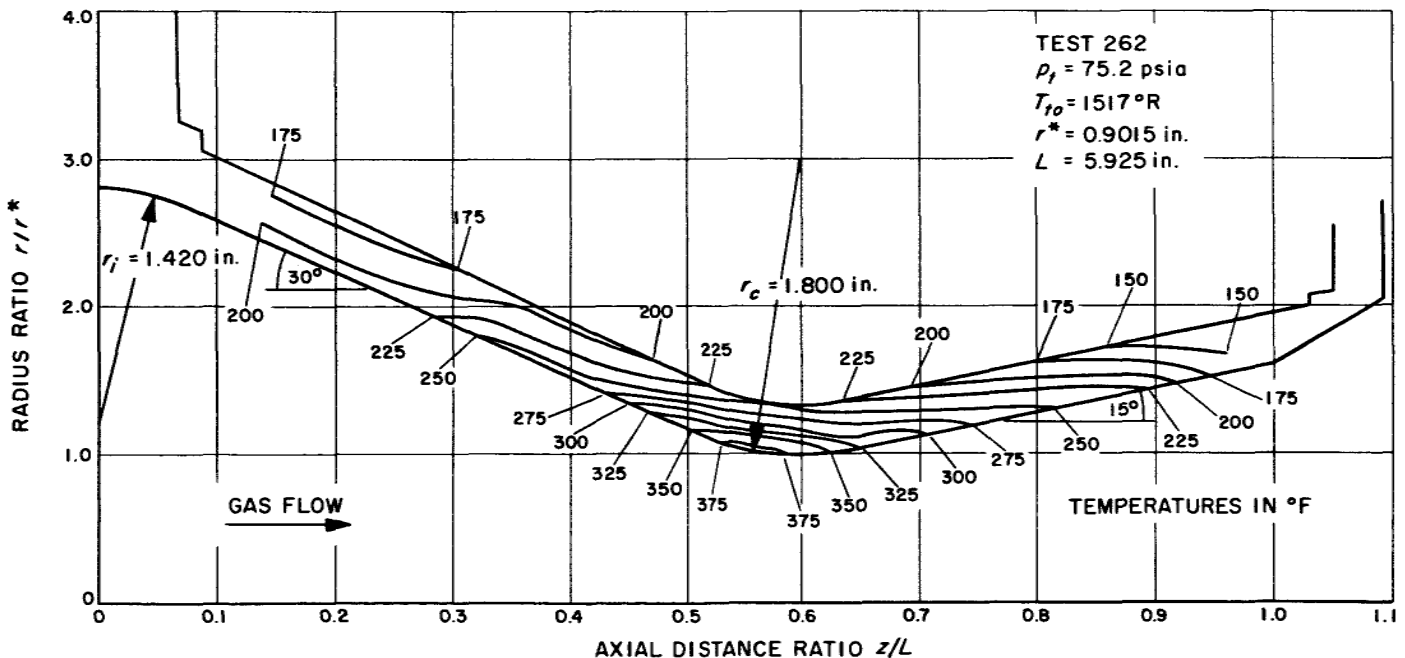


Fig. 5. Nozzle wall isotherms